

oslc4js / bmm-server MCP tool capabilities ▾

Summarize the bmm-server MCP tool capabilities

Catalogued comprehensive OSLC Business Motivation Model server capabilities >

Your `bmm-server` exposes a clean OSLC interface over the OMG Business Motivation Model. The 22 tools fall into three functional groups.

Discovery / introspection

- `read_catalog` — returns the full ServiceProvider Catalog: every SP with its creation factories, query capabilities, resource types, vocabulary references (`oslc:domain`), and shape references (`oslc:resourceShape`). Mirrors the `oslc://catalog` MCP resource.
- `read_service_provider` — same payload as the catalog but for a single SP, fetched on demand by URL. This is the lazy-drill-down path when the catalog holds many providers and you don't want startup to crawl all of them.
- `list_resource_types` — enumerates discovered types with their factories, query capabilities, and property summaries.

Generic resource CRUD

- `get_resource` — fetch any OSLC resource by URI with all properties.
- `update_resource` — update an existing resource.
- `delete_resource` — delete by URI.
- `query_resources` — OSLC query against a `queryBase` capability URL. With the consolidated one-QueryCapability-per-SP design, you narrow by type through the filter, e.g. `oslc.where=rdf:type=<http://www.omg.org/spec/BMM#Vision>`, plus optional `orderBy` and `select` projection.

Typed creation factories (BMM domain)

`create_service_provider` provisions a new SP container (slug → `/oslc/<slug>`); note its caveat that the server must restart afterward to surface the new per-SP create/query tools. The rest are one factory per BMM concept, all sharing the common Dublin Core / OSLC property core (`title`, `description`, `identifier`, `creator`, `created`, `instanceShape`, `serviceProvider`, plus the generic link predicates `derives`, `elaborates`, `refine`, `satisfy`, `trace`, `external`, `source`):

- **Ends** — `create_visions` (`madeOperativeBy` → Mission, `amplifiedBy` → Goals), `create_goals` (`includesGoal`, `quantifiedBy` → Objectives), `create_objectives` (`measureOfProgress`, `targetDate`, `includesObjective`).
- **Means** — `create_missions`, `create_strategies` (`channelsEffortsToward`, `enablesEnd`, `includesStrategy`), `create_tactics` (`implements` → Strategy, `effectsEnforcementOfBusinessRule`, `enablesEnd`, `includesTactic`).
- **Directives** — `create_business_policies` (`governs`, `governsProcess`), `create_business_rules` (`basedOn` → Policy, `enforcementLevel`, `enforcedByBusinessProcess`, `governs`).
- **Influencers & assessment** — `create_influencers` (`influencerCategory`), `create_assessments` (`required` `assesses` → Influencer and `assessmentCategory` SWOT value; `affectsAchievementOfEnd`, `affectsEmploymentOfMeans`, `identifiesPotentialImpact`), `create_potential_impacts` (`isRiskForEnd/Mean`s, `isRewardForEnd/Mean`s, `providesImpetusFor` → Directive).
- **Org & supporting** — `create_organization_units`, `create_business_processes` (`realizes` → Tactic, `governedByBusinessRule`), `create_assets` .

The notable design points: a single consolidated QueryCapability per ServiceProvider (type filtering pushed into `oslc.where` rather than separate query bases), typed factories that enforce BMM's relationship semantics (the `assesses` / `assessmentCategory` requirement on Assessments is the only place beyond `title` that mandates fields), and the `source` / `derives` pairing that lets a

BMM element trace back to a non-OSLC origin resource — useful for the digital-thread bridging you've been doing.

One operational gotcha worth flagging: `create_service_provider`'s restart requirement means new-SP provisioning isn't a hot path; you'd batch SP creation, restart, then author resources into it.

Want me to exercise any of these against the live server — e.g. `read_catalog` to see what's currently provisioned, or `list_resource_types` to confirm the factories resolved?

What service providers are available from the bmm-server right now?

I'll pull the live catalog.

`Read_catalog` >

Right now there's a single ServiceProvider: **EU-Rent**.

- **URI:** `http://localhost:3005/oslc/eu-rent`
- **Domain:** `http://www.omg.org/spec/BMM` (one vocabulary)
- **Creation factories:** all 14 BMM types — Assessments, Assets, Business Policies, Business Processes, Business Rules, Goals, Influencers, Missions, Objectives, Organization Units, Potential Impacts, Strategies, Tactics, Visions.
- **Query capability:** one consolidated "Query BMM Resources" base at `.../oslc/eu-rent/query`, matching the one-QueryCapability-per-SP design.

A couple of things worth noting from the payload. Every factory shares the same creation URL (`.../eu-rent/resources`) and differentiates purely by `rdf:type` and shape — so creation is type-driven at the payload level rather than endpoint-driven.

The shapes all resolve to <http://localhost:3005/domain/BMM-Shapes#...>, a single shape document hosting all 14 NodeShapes. And the catalog's only declared domain is BMM, so nothing else is provisioned alongside it yet.

EU-Rent is the classic OMG worked example (the car-rental business used throughout the BMM and SBVR specs), so this looks like a reference dataset rather than your Municipal Reference Model content. Want me to run the query base to see how populated it actually is — or pull the BMM-Shapes document to check the shapes resolved cleanly?

What is EU-Rent's vision?

Query_resources >

EU-Rent's Vision is:

"Be the car rental brand of choice for business users in the countries in which we operate."

(titled more tersely as "Be the car rental brand of choice for business users")

The resource is well-connected in the BMM sense:

- It's **madeOperativeBy** one Mission ([...moeotxb1xk87gu](#)) — the Means that operationalizes the Vision.
- It's **amplifiedBy** four Goals ([...ndi27kx](#) , [...jkhe50](#) , [...pzn1x1](#) , [...ig94wa](#)) — the more specific desired results that sharpen the Vision into measurable ends.

Created 2026-04-25 by the [mcp](#) creator, so this is your AI-authored EU-Rent dataset rather than a hand-loaded one.

Want me to resolve those links — pull the Mission and the four amplifying Goals —

so you can see the top of the End/Mean hierarchy fully expanded?

What are the goals EU-Rent needs to achieve in order to realize that vision, and how are those goals quantified by concrete objectives?

Orchestrated retrieval of goals and their quantifying objectives >

I'll resolve the four amplifying Goals, then trace each one's quantifying Objectives.

Loaded tools, used a tool >

Now I'll fetch all four Goals.

Catalogued four organizational goals and their quantified objectives >

Now the Objectives. Three of the four Goals carry **quantifiedBy** links — I'll resolve those four Objectives.

Used 4 tools >

Here's the full End hierarchy under EU-Rent's Vision — the four amplifying Goals and the Objectives that quantify them.

Goal 1 — Be a premium brand car rental company *"Positioned alongside companies such as Hertz and Avis."* This Goal is the only one quantified by **two** Objectives, splitting the target by market:

- *A C Nielsen top 6 in EC countries by year-end* — be rated by A C Nielsen among the top 6 rental companies in each European Community operating country. Measure of progress: A C Nielsen quarterly rankings.
- *A C Nielsen top 9 in non-EC countries by year-end* — the same standard relaxed to top 9 in all other operating countries. Same measure.

Goal 2 — Provide industry-leading customer service

- *85% customer satisfaction score by year-end* — score 85% on the EU-Rent quarterly customer satisfaction survey. Measure of progress: the quarterly survey score.

Goal 3 — Provide well-maintained cars

- *Less than 1% mechanical breakdown rate (Q4)* — during Q4, no more than 1% of rentals require a replacement car due to mechanical breakdown (accidents excluded). Measure of progress: percentage of rentals needing a replacement for that reason.

Goal 4 — Vehicles available when and where expected *"Have vehicles available for rental when and where customers expect them."* This Goal has **no quantifiedBy link** — it currently sits unquantified, with no Objective attached.

A few observations from the trace, since this is your dataset:

The BMM modeling is textbook — every Objective carries a measureOfProgress and a time-bound target (all "by year-end" or "Q4"), which is exactly what distinguishes a BMM Objective from its parent Goal: the Goal states the *desired state*, the Objective makes it *measurable and time-boxed*. That's clean.

The one gap is Goal 4. It's amplified by the Vision but dangles with no Objective, so it can't be tracked. If this is meant to mirror the canonical OMG EU-Rent example, the spec does leave some goals less elaborated, so it may be faithful rather than incomplete — but if you're using this as a reference dataset for the Define→Instantiate→Activate framework, an unquantified Goal is a natural candidate to flag, since Layer 3 reporting would have nothing to measure against it.

Want me to draft and create an Objective for Goal 4 (e.g. a fleet-availability / no-show-rate target) and wire up the quantifiedBy link, or pull the Mission and its Strategies/Tactics next to see the Means side that's supposed to operationalize all

this?

How does EU-Rent intend to achieve these goals and objectives?

Traced organizational means through mission, strategies, and tactics >

The "how" lives on the Means side — the Mission, and the Strategies and Tactics beneath it. Let me pull the Mission first, then query all Strategies and Tactics.

Queried tactics to complete means hierarchy >

Now the Tactics, which implement the Strategies.

Mapped strategy-to-tactic chains across organizational goals >

The Means side resolves cleanly. At the top sits the **Mission** — *"Provide car rental service across Europe and North America for both business and personal customers"* — which is what makes the Vision operative. Beneath it, Strategies channel effort toward the Goals, and Tactics implement those Strategies. Organized by the Goal each serves:

Goal 1 — Be a premium brand car rental company Two Strategies channel toward this Goal (and toward the Vision directly):

- *Nationwide on-airport head-to-head competition* — operate nationwide in each country, focusing on major airports and competing on-airport with the other premium players. Implemented by one Tactic: *encourage rental extensions to maximize utilization*.
- *Asset-light secondary-airport coverage* — extend reach to secondary airports without diluting the brand, using delivery/collection, the existing agency network, and budget-airline partnerships, funded via EC-Lease and piloted in Eastern Europe. Implemented by two Tactics: *car delivery and collection at*

secondary airports and extend the agency network near secondary airports.

Goal 2 — Provide industry-leading customer service Two Strategies:

- *Outsource loyalty rewards to a third-party scheme* — join an established rewards programme rather than building one. (No Tactics attached.)
- *Build direct customer-service capability* — develop service capability in-house rather than relying on the outsourced scheme alone, targeting the two assessed weaknesses. Implemented by two Tactics: *build a corporate self-service rental system* and *counter-staff retention with experienced cover*.

Goal 3 — Provide well-maintained cars (and Goal 4 — Vehicles available when and where expected) One Strategy serves both:

- *Manage car purchase and disposal at local area level* — local management with national guidance on models, fleet mix, disposal by mileage/age, and purchase phasing. This is the workhorse Strategy, implemented by four Tactics: *outsource maintenance for small branches*, *create standard car-model specifications*, *equalize car usage across rentals*, and *comply with manufacturers' maintenance schedules*.

A few things worth flagging, since this is your dataset and the structure tells a story:

The **provenance split is visible in the data**. The original Mission, three Strategies, and five Tactics were all authored by `mcp` on 2026-04-25 — the canonical OMG EU-Rent example. Four resources carry today's timestamps and dual `anonymous` / `mcp` creators: the two asset-light Strategies/Tactics and the customer-service ones. Those are clearly your recent extensions, and they're better-instrumented than the originals — they carry `derives` links back to Influencer Assessments and `enablesEnd` links, where the April resources mostly relied on `channelsEffortsToward`.

The *Build direct customer-service capability* Strategy is notable: its own description states it addresses "the gap between the 85% satisfaction objective and the means currently in place." So someone (you, or the agent) already spotted that the

outsourced-rewards Strategy alone couldn't move the 85% satisfaction Objective, and added a directly-managed Strategy plus two concrete Tactics to close it. That's exactly the kind of gap-closing the BMM is meant to make visible.

Two genuine gaps remain. The *Outsource loyalty rewards* Strategy has **no Tactics** — nothing operationalizes it. And **Goal 4 (vehicles available)** is enabled by the car-purchase Strategy but, as we saw last turn, still has **no Objective** quantifying it — so there's a Means pointed at it but no measurable target to confirm it's working.

Want me to close either gap — draft an Objective for Goal 4 and a Tactic or two under the loyalty-rewards Strategy — or pull the Influencers and their SWOT Assessments next, so you can see the analysis those two new Strategies were derived from?

Can you create a SWOT chart for EU-Rent?

Devised data-driven SWOT analysis from live server assessments >

A SWOT chart maps directly onto the Assessment resources — each carries an `assessmentCategory` (Strength/Weakness/Opportunity/Threat) and assesses an Influencer. Let me pull the live Assessments and the Influencers behind them so the chart reflects what's actually in the server.

Architected four-quadrant SWOT matrix with color-coded categories >

The SWOT lives in six Assessment resources — one Strength, two Weaknesses, two Opportunities, one Threat. Let me build the chart from exactly that data. Here's EU-Rent's SWOT, built from the six Assessment resources currently in the server — one Strength, two Weaknesses, two Opportunities, one Threat:



Each quadrant is clickable if you want to drill into a category. A few things worth noting about what the data does and doesn't show:

The matrix is **unevenly populated** — and meaningfully so. EU-Rent has exactly one catalogued Strength against two Weaknesses, which tilts the internal picture pessimistic. That's faithful to the canonical OMG example, but if you're using this as a live planning dataset it's the kind of imbalance worth questioning: a real premium operator almost certainly has more than one strength (brand, fleet quality, the on-airport positioning itself), they just haven't been entered as Assessments yet.

The links back to the rest of the model are the more useful part. Three of these Assessments carry `affectsAchievementOfEnd` pointing at the *premium brand* Goal

(the branch-distribution Strength, the premium-market Opportunity, and the budget-airline Threat), and the two Weaknesses point at the *customer-service* Goal. So the SWOT isn't free-floating — it's wired to the Ends it bears on. That's also exactly why those two new customer-service Strategies appeared today: both Weaknesses feed the same Goal, and the *Build direct customer-service capability* Strategy **derives** from them.

Two of the Opportunities and the premium-market one also carry **identifiesPotentialImpact** links, so there's a risk/reward layer underneath this that the chart doesn't surface.

Want me to pull the Influencers each Assessment sits on (the SWOT here is the *judgment*; the Influencer is the underlying *factor*), or render the Assessment-to-Goal wiring as a diagram so you can see which Ends are under the most external pressure?

What is EU-Rent's biggest risk and how should it be mitigated?

Excavated risk data to identify EU-Rent's primary threat >

The risk layer lives in the PotentialImpact resources — those carry the **isRiskForEnd** / **isRiskForMeans** links that formally mark something as a risk. Let me pull them.

Navigated timeout issues while synthesizing available assessment data >

The query to the live server timed out. Let me retry once — it may have been transient.

Synthesized available data to identify threat and mitigation strategy >

Two calls to the live server timed out, so I can't pull the `PotentialImpact` objects right now — those are what would let me compare formal risk *magnitudes* across the model. But the answer is well-supported by what we've already retrieved this session, and the most actionable risk doesn't actually depend on that missing layer.

The biggest risk: budget airlines flying to secondary airports.

It's the standout for three reasons that are all visible in the data:

It's the *only* Threat in the entire SWOT — the sole assessment classified External · Harmful. Everything else negative is an internal Weakness, which EU-Rent controls directly. A threat is, by definition, the harder class to manage because the cause sits outside the business.

It attacks the highest-stakes End. The budget-airline Assessment's `affectsAchievementOfEnd` points at the *premium-brand* Goal — and that Goal is the most heavily-supported one in the whole model: it's amplified into the Vision, quantified by two Objectives (the A C Nielsen top-6/top-9 rankings), and targeted by more Strategies than any other Goal. So the one external threat lands squarely on the part of the strategy with the most riding on it.

And the mechanism is structural, not cyclical. Budget carriers are permanently shifting traffic to airports where EU-Rent simply has no presence. Customers arriving there can't rent from EU-Rent at all — it's a coverage gap, not a price war, so it erodes the premium-positioning Goal at its foundation.

How it should be mitigated — and largely already is, in your model.

The mitigation is the *Asset-light secondary-airport coverage* Strategy, which was created today and whose `derives` link points straight back at this exact Threat assessment. That's the right BMM response: a Threat assessment spawning a Strategy that channels effort toward the threatened Goal. Its logic is sound — extend reach to secondary airports *without* building full branches there, so the premium brand isn't diluted into a discount-branch operation. It's operationalized by two concrete Tactics: car delivery and collection at those airports, and extending the

existing hotel/travel-agent agency network nearby, with budget-airline *partnerships* turning the threat's source into a channel.

If I were stress-testing that mitigation, three things stand out. First, it's asset-light by design but still cost-bearing — the model says it's funded through the EC-Lease arrangement, so the mitigation's viability is coupled to that financing holding up. Second, it's being piloted only in Eastern Europe, so it's a hedge, not yet full coverage; the threat remains live everywhere else. Third — and this is the genuine gap — there's no Objective quantifying secondary-airport penetration, so as it stands you can't actually measure whether the mitigation is working. That's the same unquantified-target pattern we saw with Goal 4.

Two caveats on the ranking itself. I'm calling this the biggest risk on the SWOT evidence — the single Threat, hitting the flagship Goal. What I *couldn't* check, because of the timeout, is whether any **PotentialImpact** resources are flagged **isRiskForMeans** against the Strategies or Tactics themselves — an execution risk could in principle outrank an environmental one. If you restart the local MCP server, I can pull those and confirm the ranking holds, and add a measurable secondary-

airport Objective to close that gap while I'm at it.

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